

Home assignment: day 6 → day 9

Deadline: 2026-07-09, 16:59.

1. Consider the recurrence equation

$$a_n = 3a_{n-1} + 5, \quad a_0 = 1.$$

- (a) Find equation for the generating function $g(t)$.
- (b) Obtain $g(t)$ in a closed form.
- (c) Extract coefficients (a_n) from $g(t)$.

2. Extract the sequence
- (a_n)
- from the generating functions

$$g(t) = \frac{1}{(1-t)^{50}}, \quad g(t) = \frac{1}{(1-3t)^{100}}.$$

3. Find the generating function of the following sequences

$$(1, 67, 67^2, 67^3, 67^4, 67^5, \dots), \quad \left(\binom{5}{0}, \binom{6}{1}, \binom{7}{2}, \binom{8}{3}, \dots \right).$$

Hint: you don't need to solve equations, just recognize the patterns we have already seen.

4. You know that the generating function
- $g(t)$
- of some sequence
- $(a_0, a_1, a_2, \dots)$
- satisfies the equation

$$g(t) = 4 - 9t + (5t - 6t^2)g(t).$$

- (a) Find $g(t)$.
- (b) By simply plugging some value of t inside $g(t)$ find the sum

$$\sum_{n=0}^{\infty} \frac{a_n}{100^n}.$$

Demo for the quiz on day 7

1. Extract the sequence
- (a_n)
- from the generating functions:

$$g(t) = \frac{1}{1-3t}, \quad g(t) = \frac{1}{1+7t}, \quad g(t) = \frac{t}{1-3t}.$$

2. Decompose the fraction into a sum of more simple fractions

$$g(t) = \frac{4-9t}{(1-2t)(1-3t)}.$$